M383C

Chapter 1: Preliminaries

1.1: Elementary Topology

Metric Space

Topological Space

Definition

Properties

(Thm) Clopen intersection of
open and closed

Base

Definition

Properties (Thm) Base for a T

(Thm) Metric induces Topology

Closure

Definition

Properties

(Prop) Closure = Smallest
closed set containing A

Interior

Definition

Properties

(Prop) Interior = Largest open
set contained in A

(Prop) Operations on Cl and Int

Accumulation Point

Definition

Properties

Dense

Boundary

Definition

Properties (Prop) cl(A)=Int(A)U bd(A)

Convergence

Definition

Properties (Prop) x \in ac({x_n})

Hausdorff Space

Definition

Properties

(Prop) (X,T) Hausdorff =>
singleton set is closed and limit
is unique

Strict limit point

Definition

Properties

(Prop) Metric => bd(A) iso or
s.l.p.

(Prop) Metric => x_n \to x iff def.

(Prop) Metric space <=>
Hausdorff

(Prop) Metric A ac => x_n \to x

Continuous function

Definition

Properties

(Prop) composition cts

(Prop) f(x_n) to f(x)

(Prop) f seq. cts + X metric => f
cts

Homeomorphism

Definition

Properties

(Prop) S stronger than T iff
Identity cts

(Prop) weakest topology exists

Product Topology

Definition

Properties

(Prop) pi_a cts

(Prop) X, Y_a => f: X -> \Pi Y_a
is cts iff pi_a * f cts for each

Compactness

Definition

Properties

(Prop) Closed sub of compact is
compact; compact sub of Haus
if closed

(Prop) cts image of a compact
set is compact

(Prop) Metric => compact iff
seq. compact

(Thm) Tychonoff

1.2: Lebesgue Measure and
Integration

sig-alg

Definition

Properties (Prop) Set operations

Measure

Definition

Properties (Prop) Basics

Measure Space

Borel sig-alg

Lebesgue Measure on B(R^n)

(Thm) Existence of Complete
Measure Space

Measurable Functions

Definition

Properties

(Thm) 1. +-*/; 2. Composition; 3.
cts=>meas; 4. inf sup liminf
limsup

(postThm) f meas => f+, f-, |f|
meas

Characteristic and Simple
Functions

Definition

Properties (Thm) Approximation of any f

Integrable Functions

Construction

Definition

Properties

(Prop) int |f| < infty iff integrable

(Thm) Riemann Integrable on K
<=> Lebesgue Integrable on K

(Prop) Properties of meas f

(Thm) int_{A} f = \sum
\int_{A_n} f

(Thm) int on a.e.

(Thm) f bdd on K => f Riemann
Int on K iff f in cts on all but
count points

(Thm) abs. cts of lebesgue f

Important Theorems

(Thm) Lusin

(Thm) Egorov

(Thm) Chebyshev Inequality

(Thm) MCT

(Thm) Fatou

(Thm) DCT

(Thm) Fubini

(Thm) FTC

1.3: Complex Contour
Integration NOT USED YET

Chapter 2: Normed Linear
Spaces and Banach Spaces

2.1: Basics

Vector Spaces

Normed Linear Space

Definition

Examples

Properties

(Prop) (X,||.||) NLS V subspace
=> (V,||.||) NLS

(Prop) ||.|| is cts

(Prop) Every Cauchy in NLS is
bdd

Banach Space

Definition

Examples

Norm Equivalence

Definition

Properties Finite Dimension all Equivalent

Convex

Definition

Properties

(Prop) (X,||.||) NLS r>0 => B_r is
convex

2.2: Linear Maps and The Dual
Spaces

Operators

Definition

Properties

(Prop) T:X—>Y linear => T bdd
iff ||Tx|| \leq C||x||

Bounded Linear Operator

Definition

Examples

Operator Norm

Properties

(Prop) T: X—>Y linear: 1. T cts;
2. T cts at pts; 3. T bdd

(Prop) (B(X,Y), ||.||) is NLS; is
Banach <= Y Banach

Bounded Linear Functional

2.3: Important Examples

Finite Dimensional Spaces

Definition

(Thm) Young’s Inequality

(Thm) Hölder’s Inequality

(postThm) Minkovski’s
Inequality

Properties

(Prop) p in [0,\infty], in F^d => |
^infty \equiv l^p

(Prop) dim(X)< \infty => all
norms equivalent; compact =
bdd+closed

(Prop) dim(X)=d => X \iso F^d
with Homeomorphic

(Prop) dim(X)=d => 1.T bdd; 2.
B(X,F) \iso F^d with
Homeomorphic

l^p

Definition

(Thm) Hölder’s Inequality for l^p

Hamel / Schauder Basis

Definition

Example

Separable

Completeness of l^1 in \infty
norm

Properties

(Prop) p<1 => ||.||_{l^p} is not a
norm on l^p

p \in [0,\infty] => Y:l^p —> F in
B(l^p, F)

L^p

Definition

Equivalence Class & L^p

esssup/essbdd

(Thm) Hölder’s Inequality for
L^p

(Thm) Minkowski’s Inequality

Properties

(Prop) p\in (0,\infty],
L^p(\Omega) is VS and ||f||_p=0
<=> f=0 a.e.

(Prop) 1 \leq p \leq q \leq \infty.
L^q \subset L^p

(Prop) p\in [1,\infty], \Omega
meas. => L^p(\Omega) Banach

(postThm) p\in [1,\infty),
\Omega meas. f_n in L^p =>
f_{n_j} -> f \in L^p a.e. ptw.

(Prop) p\in [1,\infty), \Omega
meas. => (L^p(\Omega))^* =
{T_g: g\in L^q(\Omega)}

(Prop) p\in [1,\infty),
cl(S_0)=L^p(\Omega)

(Prop)p\in [1,\infty),
cl(C_0(\Omega))=L^p(\Omega)

2.4: Hahn-Banach Theorems

Definitions

sublinear/seminorm/norm

Ordering

(preThm) one-step Hahn-
Banach

(preThm) Zorn’s Lemma

(Thm) Hahn-Banach for R-VS

(Thm) Hahn-Banach for C-VS

(postThm) Hahn-Banach for
NLS

(Thm) Krein-Milman

2.5: Applications of Hahn-
Banach Theorem

Definition

Separable

Minkowski Functional

Properties

(Prop) X NLS => X^* separates
pts

(Prop) X NLS, X^* separable =>
X separable

(preThm) Mazur Separation
Lemma I

(preThm) Mazur Separation
Lemma II

(preThm) Separating
Hyperplane Thm

Corollaries

(postThm) X NLS, x_0 \neq 0
=> f\in X^* st ||f||=1 and f(x_0)=||
x_0||

(postThm) X NLS, x_0 \in X =>
f\in X^* st f(x_0)=||f|| ||x_0||

(postThm) Dual
Characterization of ||x||

(postThm) X NLS x\in X st
f(x)=0 for all f\in X^* => x=0

2.6: The Open Mapping
Theorem

Definitions

Nowhere Dense/First Category/
Second Category

Category Preliminary

(Thm) Baire Category Theorem

(postThm) \Bigcap of countable
open dense of complete metric
spaces is non-empty

(postThm) Second Category is
complete metric space

(Thm) Open Mapping Theorem

Theorem

(postThm) X,Y Banach, T bdd
linear bijection => T^-1 cts

Isomorphic as Banach Spaces

Definition

Example

(Thm) Closed Graph Theorem

(Prop) X Topo, Y Hausdorff, f
cts => graph(f) closed

Closed Operator

(Thm) Closed Graph Thm

(postThm) D\subset X, Y
Banach, T closed linear =>
closed D iff T bdd

2.7: Uniform Boundedness
Principle

Theorem

Proof by Baire

(preThm) X,Y NLS T\in B(X,Y)
=> sup_{B_r(x)} ||Tz||\geq ||T||r

Proof Original

2.8: The Embedding of X into its
Double Dual X^**

Evaluation Map

Isometry

Reflexive

2.9: Compactness and Weak
Convergence in an NLS

2.9.1: The Norm on Strong
Topology

(preThm) X NLS, Y closed sub.
Z sub. Y, => z \in B_1 d(z,Y)
\geq \theta

(postThm) X NLS, dim(X)=\infty,
M closed bdd nonempty int =>
M not compact

2.9.2: The Weak and Weak-*
Topology

Weak/Weak-* Convergence

(Prop) Strong => Weak

(Prop) X NLS, {x_n} in X, {f_n}
in X^* => 1. {x_n} conv weak =>
lim unique {||x_n||} bdd; 2. {f_n}
conv weak-* => lim unique (X
Banach => {||f_n||} bdd)

(Prop) X NLS, {x_n}. x_n -w-> x
=> ||x|| \leq liminf ||x_n||

Weak Topology/ Weak-*
Topology/

(Prop) Weak/Weak-*
Convergence <=> Convergence
in Weak/Weak-* Topologies

(Thm) Banach-Alaoglu

(Thm) X Separable NLS, K
\subset X^*. K weak-* compact
=> K metrizable in weak-*
Topology

Theorem

(postThm) X Separable NLS,
{f_n} \subset X^*, R>0 ||f_n||\leq
R for all n => {f_{n_i}}
converges weak-* in X^*

(Thm) Generalized Heine-Borel
I

Theorem

(postThm) X Banach,
Separable, Reflexive, {x_n} bdd
=> {x_{n_i}} converges weakly
in X

Example on L^p

(Thm) Eberlein-Šmulian

(Thm) Generalized Heine-Borel
II

(Thm) Banach-Saks

Theorem

(postThm) X NLS K nonempty,
convex, closed => {x_n}
converges weakly to x\in K

2.10: The Dual of an Operator

Dual/Conjugate/Adjoint
Operator

Definition

3Properties

(Prop) 1. Norm Preserve; 2.
Linear; 3.Transpose; 4. Id Dual

Examples

Matrix

Dual L^p on Kernel

(preThm) X,Y NLS, T, => T^**
extension of T. Reflexing =>
T^**=T

(preThm) X,Y Banach, T => bdd
inverse T <=> bdd inverse T^*

Chapter 3: Hilbert Spaces

3.1: Basic Properties of Inner
Products

Definition

IPS

Basics

(Prop) Second Arg Conjugate
Linear + Sesquilinear

Examples

F^d

l^2

L^2(\Omega)

(preThm) Cauchy-Schwarz

(postThm) IPS\subset NLS

Orthogonal

Properties

(Prop) Polarization

(Prop) Parallelogram +
Pythagorean

(preThm) H IPS => <,> cts

(postThm) <a_n x_n, b_n y_n>
— > <ax,by>

3.2: Best Approximation and
Orthogonal Projection

(Thm) Best Approximation
Theorem

Theorem

(postThm) H IPS, M complete
subspace => <x-y, M>=0

Orthogonal Complement

(Prop) H IPS, M\subset H =>
1.M^⊥ closed; 2. M ⊥ M^⊥; 3.
M\bigcap M^⊥={0} or \empty

Idempotent/Projection

(Thm) X NLS, P Projection onto
M, N = KerP=> 1.P=I on M,
2.Q=I-P Projection onto N,
3.QP=PQ=0, 4.X=M\dir+N, 5.P
bdd, M≠{0} => ||P||≥1

Theorem

Projection (Oblique) => Spaces

(Thm) H IPS, M complete
subspace => ! P, P^⊥ 1. bests,
2. linear direct, 3. y=Px <=> x-
y⊥m, 4. <Px,P^⊥x>=0, 5. Pyth,
6. ||P||=1 if M≠{0}, ||P^⊥||=1 if
M≠N

Theorem

Spaces => Projections (⊥)

Corollary

(postThm) 7.PP^ ⊥=P^ ⊥P=0,
Idempotents, 8.Cross 0,
9.Cross 0

(postThm) H Hilbert, M
complete subspace =>
P^⊥_M=P_{M^⊥}

3.3: Duality in Hilbert Spaces

Norm Preserving on H^*

(Thm) Riesz Representation

Riesz Map/Hilbert Adjoint

(Prop) X,Y Hilbert, T\in B(X,Y)
=> T^*\in B(Y,X), T^**=T

3.4: Orthonormal Subsets and
Bases

ONS

Linear Independence

(Prop) A\subset H Hilbert
orthogonal => A linear
indepedent

(Thm) H+cONS, , M =
span{u_a} => 1. P_M = \sum
<.,u_a>u_a 2. countable Bessel

(Thm) Uncountable Bessel

Uncountable Index Set

Theorem

(postThm) At most Countable
<x,u_a>

l^2(I)

(Thm) Riesz-Fischer

(Thm) H+ONS, TFAE 1. {u_a}
maximal ONS, 2.{u_a} ONB,
3.Parseval on ||x||^2, 4.Parseval
on <x,y>

(Thm) H+ONS, Riesz-Fischer
Map is essentially countable

(postThm) H+ONS, Riesz-
Fischer Map is Hilbert Isom

(Thm) H+ONS, {u_a}\subset
{u_b} ONB by Gram-Schdmit

(postThm) Every Hilbert H is
\iso l^2(I) for some I. H d-\iso
l(N) dim(N)=\infty separable

(Thm) {e^{inx}}_{n=-\infty}
^{\infty} is dense in L_{per}^2(-
\pi,\pi)

3.5: Weak Convergence in
Hilbert Space

(preThm) H+ONB => x_n -w-> x
<=> ||x_n|| + <x_n,e_a> —>
<x,e_a>

(preThm) H Separable, {x_n}
bdd => {x_{n_j}} weakly
convergent

L^2_per(-\pi,pi)

Chapter 5: Distributions

5.1: The Notion of Generalized
Functions

Motivation

Problem

IBP

Solution

Example

5.2: Test Functions

supp/\alpha/C^m/C^\infty/D/D_K

(Prop) C^m C^\infty D D_K are
nontrivial VS

(postThm) there is C^\infty non-
analytic f

Convergence in D

Definition

Counter

(Thm) (D(\Omega), T_{K,p}) is
complete

5.3: Distributions

Distribution

cts<=>cts at 0<=>K-bdd

(Thm) T: D(\Omega)—>Y linear,
is cts <=> seq cts

(Thm) T: D(\Omega)—>F linear
=> T seq. cts <=> T seq. cts at 0

(Thm) T: D(\Omega)—>F linear.
T \in D’ (cts) <=> K-bdd

L^1_loc(\Omega)

(preThm) Lebesgue Lemma

Regular/Singular Distributions

Motivation from L^1_loc

Definition

Dirac Delta

Examples

5.4: Operations with
Distributions

Setting

(Prop) T: D(T)—>D(T) linear +
cts => T*u \in D’(\Omega)

Multiplication with Smooth
Functions

Definition

Regular case

Differentiation

Definition

(Prop) u \in C^n, D^a u = D^a
\Gamma_u = \partial^a u =
\Gamma_{\partial^a u}

Prop

Example

(Prop) u \in D’ => D^a D^b u =
D^b D^a u = D^{a+b} u

(preThm) Leibniz Rule

Theorem

Example

Translations and Dilations of
R^n

Definition

Distributional

Convolutions

Definition

Classical

Distributional

Example

(Prop) 1. t_x(u*\phi) = (t_x
u)*\phi = u (t_x * \phi) 2. u*\phi
\in C^\inf, D^a (u*\phi) = (D^a
u)*\phi = u * (D^a \phi)

Distributivity of t_x and D^a

Proposition

phi, psi \in D => C^\inf_0

(Prop) \psi, \phi \in D, u\in D’ =>
(u*\phi)*\psi = u*(\phi*\psi)

Associativity D’ * D * D

Proposition

5.5: Convergence and
Approximations to Identity

Topology of D

D reflexive

Notion of Convergence

(Thm) Banach-Steinhaus

(Prop) {u_n}, {<u_n,\phi>}
cauchy => u(\phi)=<u,
\phi>=lim<u_n,\phi> in D’

(preThm) T: D—>D linear cts,
u_n—>u in D’ => T^*u_n—
>T^*u

Lemma

(postThm) u_n—>u in D’ =>
D^au_n —> D^au in D’

(Prop) lim 1/h (u-t_ha u)=D^au
in D’

(Prop) 1/e X_{[-e/2,e/2]}—>d_0

Approximation to Identity

(Prop) {\phi_e} => \int \phi_e =1
+ supp\phi_e —> {0}

(Thm) 1. \psi * \phi_e —> \psi;
2. u* \phi_e —> u; 3. u*\phi_e
\in C^\infty dense in D’

(postThm) \phi_e = d_0 * \phi_e
— > d_0

5.6: Some Applications to
Linear Differential Equations

Classical/Weak/Distributional

ODE

d=1

(preThm) \phi \in D + \int \phi =0
=> \psi’ = \phi, \psi \in D

(Thm) u\in D’ has infty many of
primitives differ by constant

Theorem

(postThm) u’=0 => u = C

(postThm) a\in C^\infty => u’=
au = 0 gives u = Ce^{ax}

(postThm) a,b\in C^\infty => u’+
au = b gives u = e^-A [\int_0^x
e^Ab +C]

Example xu’=1

PDE

d>1

Wave Operator

Fundamental Solution

(Thm) Lu=f => u = E*f

(Thm) Malgrange Ehrenpreis

Wave

Laplace

(Thm) f, E*f \in L^1 => E*f \in
L^1_{loc}

(Thm) Structure Theorem 5.7: Local Structure of D’

Chapter 4: Spectral Theory and
Compact Operators

4.1: Definitions of \rho, \sigma

Motivation

Resolvent Set and Operator

Spectrum

(Prop) 1. \sig=C\ \rho=\sig_p U
\sig_c U \sig_r; 2. \lambda ≠
0=> N(T_\lambda)≠{0}

Eigenvalues/Eigenfunctions

Definition

Examples 1.Shift 2. D=‘

4.2: Basic Spectral Theory in
Banach

(preThm) X Banach, T \in B(X),
\lambda \in \rho => T_\lambda
maps onto X

(postThm) X Banach, T \in B(X)
=> \lambda \in \rho <=>
T_\lambda invertible on X

(preThm) Neumann Series

(postThm) General Neumann
Series

(postThm) X Banach => GL(X)
is Open in B(X)

(postThm) X Banach, T \in B(X)
=> 1. \rho is Open, 2. \sigma is
compact, 3. |\lambda| ≤ ||T|| in
\sigma

4.3: Compact Operators on a
Banach Space

Compact Linear Operator

NLS Series [BCC, 3AE, WcaoF]

(Prop) X,Y NLS, T \in C(X,Y) =>
T bdd (cts)

(preThm) (X,d) Metric => X
compact <=> X seq compact

(preThm) X,Y NLS, T linear =>
T Compact with seq
characterization

(Prop) X,Y NLS, T linear => 1.
dim(X)<\infty => T compact, 2. T
bdd, dim(Y) <\infty => T
compact, 3. dim(X)=\infty => I
not compact

(Thm) X NLS, Y Banach, {T_n}
\subset C(X), ||T_n -T|| —>0 =>
T \in C(X)

Example

(Thm) X,Y NLS, T\in C(X,Y),
x_n -w-> x => Tx_n —> Tx

(Prop) X NLS, T\in C(X) => 1.
\sigma_p countable, 2. infinitely
many => accumulate at 0 only,
3.dim(X)=\infty => 0\in \sigma_p

CA0

Theorem

(Prop) X NLS, T\in C(X) =>
\lambda ≠ 0,
dim(N(T_\lambda)) <\infty

Banach Series [bir R^n p S F]

(preThm) X,Y Banach, T\in
B(X,Y), ||Tx||≥\gamma ||x|| => T
injective, Ran(T) closed

(Prop) X Banach, T\in C(X) =>
\lambda ≠ 0, R(T_\lambda^n)
closed

(Thm) X Banach, T\in C(X),
\lambda \in \sigma(T) and
\lambda≠0 => \lambda \in
\sigma_p

(Thm) Spectral Theorem for
Compact Operators

Theorem

C+A0F

(Thm) Fredholm Alternative for
Compact Operators

4.4: Bounded Self-Adjoint
Linear Operators on Hilbert
Space

Self-Adjoint

(Prop) H Hilbert, T\in B(H,H) 1.
T=T^* => <Tx,x>\in R 2. H
complex => T=T^* <=> <Tx,x>in
R

(preThm) Bdd below => 1. T 1-
to-1, 2. Ran(T) closed

(Thm) Spectral Theorem for
Self-Adjoint I

Rayleigh Quotient

(Thm) Spectral Theorem for
Self-Adjoint II

Positive Operator

Definition

(Prop) H complex Hilbert, T\in
B(H,H) => T postive <=> self-
adjoint \sig(T)≥0

(Prop) T+S, T^n, T^-1, aT

Square Root of T

Definition

(Thm) H complex Hilbert =>
1.T\in B(H,H) unique sqrt 2.
ST=TS

Example

4.5: Compact Self-Adjoint
Linear Operators on Hilbert
Space

(Thm) Hilbert-Schmidt

(Thm) Spectral Theorem for
Compact Self-Adjoint

(Prop) Simultaneously
Diagonalization

4.6: The Ascoli-Arzelà Theorem

Main Goal

(preThm) Compact Metric
Space is Separable

(Thm) Ascoli-Arzelà

(Thm) Integral Operator with
smooth kernel on \Omega is
Compact

(postThm) Integral Operator
with L^2 kernel on \Omega is
Compact

4.7: Sturm-Liouville Theory

Second Order Setting

L: C^2 \to C^0

(Thm) Picard

(postThm) dim(N(L))=2

Formal Adjoint/Formally Self-
Adjoint/Real

(Prop) real L, a_1=a_0’ <=> L
self-adjoint

(Prop) Integrating Q to make L
self-adjoint

Examples [LC/LH/B]

4.7.1: SLP and Green

Regular SLP

Definition

Solution Operator

Green’s Function

Definition

Example

(preThm) Abel’s Thm

(preThm) W≠0 => u\perp v

(Thm) Regular SLP

4.7.2: Spectral Properties of
Solution Operator

Solution Operator

T

Bdd

<,>_{L^2}~<,>_{w}

Self-Adjoint in (C^0, <,>_w)

(preThm) T: L^2(I)—> C^0(I) is
compact, takes C^0 —> C^2

Spectral Properties of T

(Prop) 0 \notin \sig_p(A) => 0
\notin \sig_p(T)

(Prop) \lambda ≠ 0 => \lambda
\in \sig_p(A)<=>1/\lambda \in
\sig_p(T)

(Prop) \sig_p(A) is simple

(Thm) Summary

4.7.3: Some Applications

Intuition

Dirichlet/Fourier1/
Fourier2/\Omega/Neumann/
Periodic

Eigenfunction Expansion